PORTFOLIO · 2026

Service Planner

The workshop day on one board: every car, every stage, every technician.



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Workshop planner for a multi-brand premium service centre · 5 brands, 5 roles

THE BRIEF

What the board had to guarantee

Type the car once

The advisor enters the order at reception. Every later stage inherits it instead of retyping it.

No release without QC

A car cannot be handed back to its owner until quality control has signed it off.

Nothing falls off overnight

A job that runs into tomorrow continues on tomorrow's board and keeps its history.

HOW IT WORKS

Four stages, one order

01

Advisor

Reception

Order, plate, VIN and complaint, typed once. Job, QC and handover are created with it.

02

Technician

Job

Dragged into a technician's column and time slot. Resized to its real length.

03

Quality control

QC

Its own stage with its own sign-off. Until it is done, handover stays locked.

04

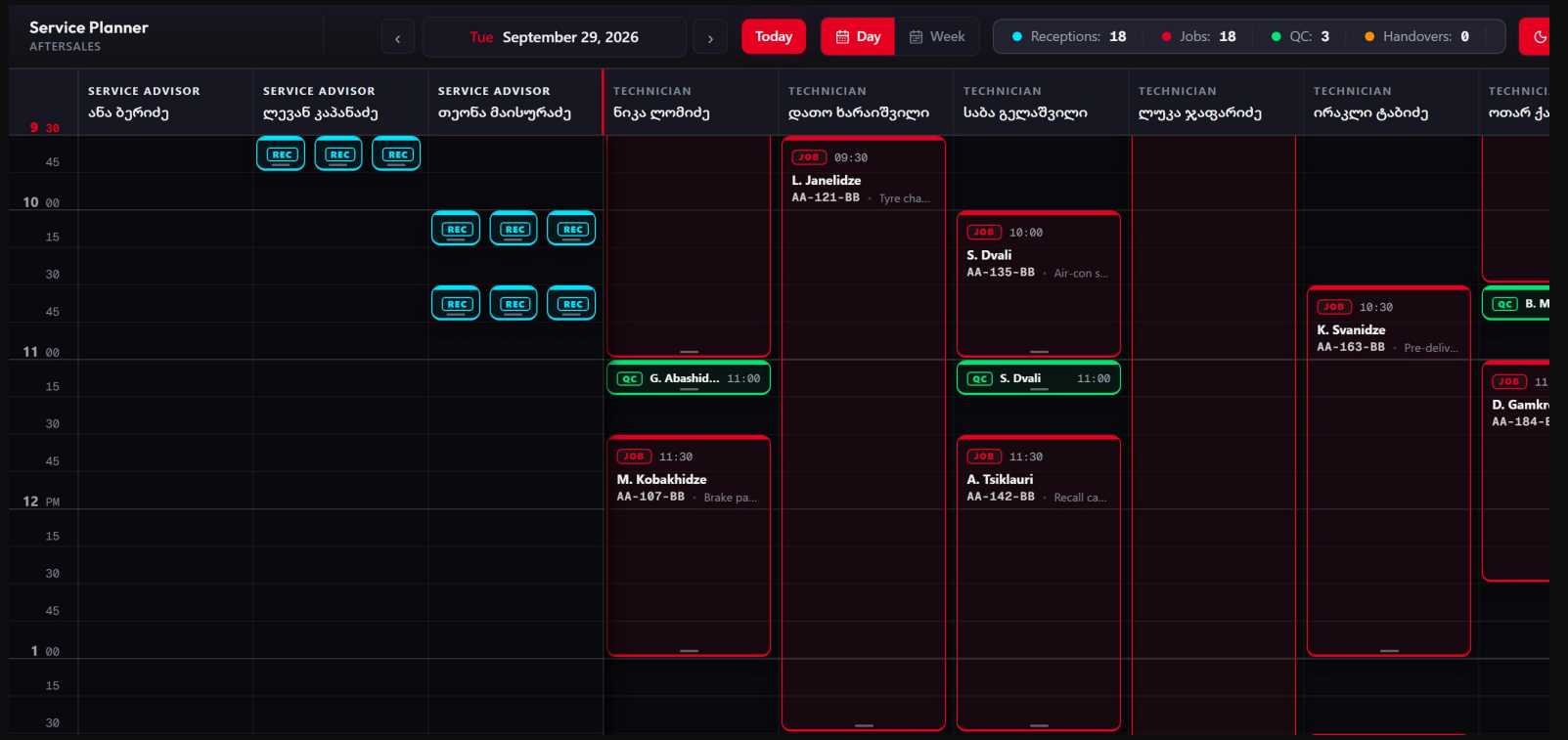
Advisor

Handover

Completing it closes the whole order and notifies the team.

HOW IT WORKS

The day board



Live across screens

A change on one screen reaches every other in about a second.

Built for the real shop

One technician, two cars: the cards sit side by side instead of hiding each other.

Every monitor

The header scales to fit instead of dropping controls, even at 125 % Windows scaling.

Sample data · made-up names

THE RULES

Rules the server enforces

The QC gate

Handover cannot complete while the order's QC stage is open. Verified end to end: blocked, sign off QC, then allowed.

Carry-over is a chain

Tomorrow's continuation links back to today's card. Never a copy, so nothing is counted twice.

Parking keeps the stage

A car waiting weeks for parts is parked with its own flag. Its stage, and everything counted on it, stays put.

Brands are walled off

Each account is granted brands explicitly. No grant means no access, not all access.

THE KEY DECISION

One field everything counts on must never be reused for anything else.

1

field: the stage

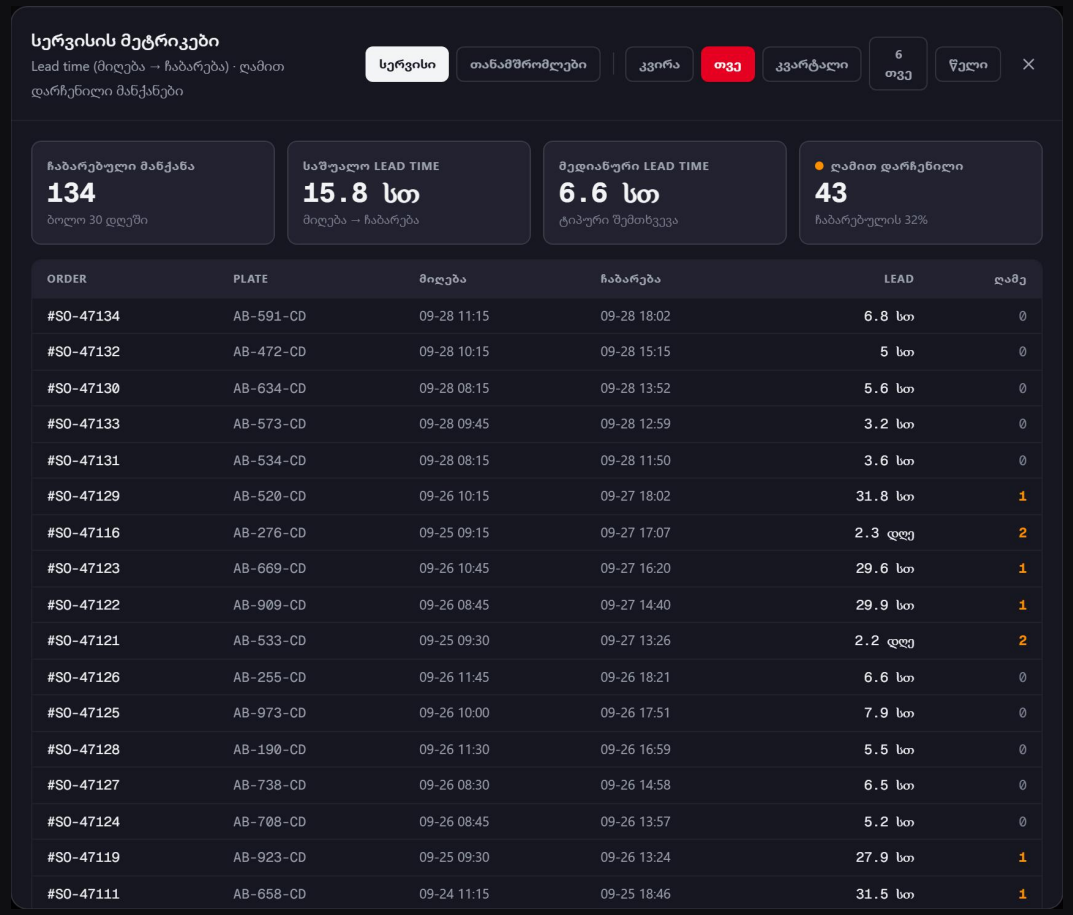
6

things counted on it

Technician load, the wall display, the QC gate, carry-over, order closing and the statistics all read the stage. Parking a car used to overwrite it, and the car vanished from all six. Now parking is its own flag and the stage is never touched.

WHAT MANAGEMENT READS

Lead time, reception to handover



ORDER	PLATE	მიღება	ჩაბარება	LEAD	ღამე
#SO-47134	AB-591-CD	09-28 11:15	09-28 18:02	6.8 სთ	0
#SO-47132	AB-472-CD	09-28 10:15	09-28 15:15	5 სთ	0
#SO-47130	AB-634-CD	09-28 08:15	09-28 13:52	5.6 სთ	0
#SO-47133	AB-573-CD	09-28 09:45	09-28 12:59	3.2 სთ	0
#SO-47131	AB-534-CD	09-28 08:15	09-28 11:50	3.6 სთ	0
#SO-47129	AB-520-CD	09-26 10:15	09-27 18:02	31.8 სთ	1
#SO-47116	AB-276-CD	09-25 09:15	09-27 17:07	2.3 დღე	2
#SO-47123	AB-669-CD	09-26 10:45	09-27 16:20	29.6 სთ	1
#SO-47122	AB-909-CD	09-26 08:45	09-27 14:40	29.9 სთ	1
#SO-47121	AB-533-CD	09-25 09:30	09-27 13:26	2.2 დღე	2
#SO-47126	AB-255-CD	09-26 11:45	09-26 18:21	6.6 სთ	0
#SO-47125	AB-973-CD	09-26 10:00	09-26 17:51	7.9 სთ	0
#SO-47128	AB-190-CD	09-26 11:30	09-26 16:59	5.5 სთ	0
#SO-47127	AB-738-CD	09-26 08:30	09-26 14:58	6.5 სთ	0
#SO-47124	AB-708-CD	09-26 08:45	09-26 13:57	5.2 სთ	0
#SO-47119	AB-923-CD	09-25 09:30	09-26 13:24	27.9 სთ	1
#SO-47111	AB-658-CD	09-24 11:15	09-25 18:46	31.5 სთ	1

Sample data

Average and median, side by side

15.8 h against 6.6 h in this sample. The overnight cars explain the difference, so they are counted too.

Nights, per order

Each carry-over link is one more night in the shop, read straight from the chain.

Per person as well

A second tab shows each staff member's load over the same window.

AROUND THE BOARD

Built for the workshop floor

Wall display

A TV in the workshop shows the day. It needs its own token and shows no customer data.

Car history by VIN

Typing a VIN brings up that car's earlier visits: dates, technicians and complaints.

Voice dictation

Technicians dictate notes in Georgian instead of typing with gloves on.

Handover alerts

Completing a handover sends a group message and an email.

Week view

Seven days at a glance, one click back to any day's board.

Phones

A horizontal timeline for small screens instead of a squeezed desktop grid.

HOW IT IS BUILT

The architecture

Browser

React and Vite, served from Cloudflare Pages. Realtime updates over a WebSocket.

API

Python, FastAPI and SQLAlchemy in one Docker image. Signed tokens checked against published keys.

PostgreSQL

Managed Postgres in the EU with row security on. SQLite for local runs.

63

automated API and role tests

165

commits

8

backend dependencies, all standard

HOW IT IS BUILT

Access and security

Five roles

Admin, senior, advisor, technician and QC. Every write asks: may this role change this card?

Brand grants fail closed

An account with no grant sees nothing. Writes take the brand from the card, not from the request.

Database locked down

Row security denies the public key everything, checked with the key itself, not assumed.

No self sign-up

Accounts are created by an admin only.

Smaller attack surface

API docs off in production, length caps on every input, a rate limit in front.

Pinned relay

The notification relay forwards for one known bot only, never for any token it is handed.

OUTCOME

It became the specification

A vendor rebuilt it

The company's main planner was rebuilt by a vendor from this code and my technical specification.

I audited their build

Side by side with this one: 40 findings, ranked by priority. 4 blocked daily use.

Handed to IT

A code package with full IT documentation, including the known issues, for running it on an internal server.

WHAT I LEARNED

Lessons from the build

Subscribed is not delivered

The realtime channel said subscribed even for a table it was not publishing. Only a real change arriving in a second tab proves it works.

Build time is not run time

A hosting screen labelled secrets only reached the build, never the running function. The fix was to stop relying on it.

Utility classes can lie

In the CSS framework's new version, move and scale are separate properties, so a transition on transform snapped silently. Check the compiled CSS, not the class names.

SUMMARY

The project in numbers

5

brands on one board

5

roles, checked on every write

63

automated tests

40

findings in the vendor audit

Python · FastAPI · SQLAlchemy · PostgreSQL · React · Vite · Realtime · Docker · Cloudflare Pages

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